A Two-Tier Retrieval-Augmented Chatbot for University Student Support Using RAG and Secure Personalisation

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Abstract

Universities require intelligent digital systems to manage a high volume of queries on courses, regulations, assessments, and other services. Traditional methods, such as help desks and frequently asked questions, are not effective in handling the increasing number of students. This project aims to mitigate the problem by developing a Two-Tier Retrieval-Augmented Generation (RAG) Chatbot System for the students of a university. In the proposed system, the architecture is divided into two layers to make the chatbot more accurate and reliable. In the proposed system, FAISS is used to enable efficient vector similarity search, BGE embeddings to enable semantic representation of documents, and Llama 3.1 to enable the generation of responses in the form of natural language. In the proposed system, Role-Based Access Control (RBAC) is integrated to make the system more secure by allowing users to have access to only the information to which they are allowed. In the proposed system, Precision@5, Recall@5, Response Latency, Access Validation Tests are performed to evaluate the performance of the proposed system. From the experimental results, it is evident that the proposed system is more accurate in retrieving academic information and is efficient in terms of response time.

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List of Abbreviations

UG	Undergraduate
PG	Postgraduate
PhD	Doctor of Philosophy
RAG	Retrieval-Augmented Generation
RBAC	Role-Based Access Control
LLM	Large Language Model
FAISS	Facebook AI Similarity Search

1 Introduction

1.1 Background

The chatbots developed using the concept of Large Language Model (LLM) have evolved significantly (Fan et al., 2024). The chatbots are now widely used by various industries to provide the facility of retrieving information. The latest advancements in generative models of artificial intelligence have allowed chatbots to understand natural language input. In the context of higher education, the chatbots may play an important role by providing students with an opportunity to access relevant information instantly. The relevant information may include the rules and regulations of the university, the course schedule, and other important rules.

The previous chatbot models used by universities were primarily based on Frequently Asked Questions (FAQs) or rule-based models. The previous models failed to provide the required support to students by interpreting complex questions. The latest LLM models are also reported to suffer from the hallucination problem, which means the chatbot may provide responses to the students' queries that may be incorrect. To overcome the limitations of the previous models, the concept of Retrieval-Augmented Generation (RAG) has been introduced (Brown et al., 2025). The RAG models use the concept of information retrieval to provide the students with the required support.

1.2 Problem Statement

Universities also have a large amount of information related to academic activities. This includes course-related information, student-related information, and staff-related information. This is crucial for the proper running of a university. Students need to retrieve this information to understand various academic activities. However, it is difficult to retrieve required information from traditional university information systems.

One of the challenges associated with university information systems is to ensure data security. The data should be protected so that only allowed students or staff can retrieve it. Another challenge is to provide accurate answers to user queries. The traditional information retrieval system is not able to provide accurate answers to students when they query the system with questions. This is because students query the system with context-based questions. Therefore, a more intelligent support system is required.

1.3 Research Aim and Objectives

The objective of this study is to build a secure two-tier Retrieval-Augmented Generation (RAG) chatbot that retrieves both publicly available information and personalised academic records while ensuring role-based access control. The proposed system is intended to enhance the accessibility and usability of academic information for various classes of users in a university setup. In the proposed system, the first tier is primarily focused on retrieving publicly available information concerning a university. The publicly available information is retrieved using various sources such as university websites and academic information. The second tier is primarily focused on retrieving personalised information concerning students, lecturers, and administrative staff. The proposed system is intended to ensure that each user is able to access information that is allowed according to their role in a university setup. In achieving the objective of the study, various objectives have been defined. These objectives include developing a public knowledge RAG system, developing a personalised information retrieval system, ensuring role-based access control, and evaluating the performance of the system using suitable evaluation criteria.

1.4 Research Questions

The research questions that guide this study include:

RQ1: How effective is a two-tier Retrieval-Augmented Generation (RAG) structure in enhancing the precision of information retrieval in university chatbot systems?

RQ2: How significantly does role-based access control (RBAC) improve the security of personalized information retrieval in chatbot systems?

RQ3: How does query routing and access control influence precision, recall, and response time in information retrieval?

RQ4: How does the performance of the proposed system compare in public (Tier 1) and personalized (Tier 2) information retrieval scenarios?

1.5 Contributions

The contribution of the research is centered around the development of a secure and efficient chatbot system for the support of university students. The first contribution of the research is the development of a two-tier Retrieval-Augmented Generation (RAG) architecture, where public information retrieval is separated from personalized academic information retrieval. The study has also contributed to the development of a secure document filtering system, where the role-based information retrieval is implemented to ensure that the user has access to the information that is allocated to their role. Moreover, the study has contributed to the development of the evaluation framework, where the efficiency of the system is evaluated based on the accuracy of the information retrieved by the system.

2 System Architecture

2.1 Overall System Architecture

The overall architecture of the proposed system follows a two-tier information retrieval model that handles both public and personalized university information based on the concept of Retrieval-Augmented Generation (RAG) (Lewis et al., 2021). The system starts with a user providing a query by interacting with the chatbot interface. The query is processed using an intent classification module, which checks whether it is a public university information query or a personalized information query. The query is then passed to either the public information retrieval model (Tier-1) or the secure personalized information retrieval model (Tier-2), depending on the classification outcome.

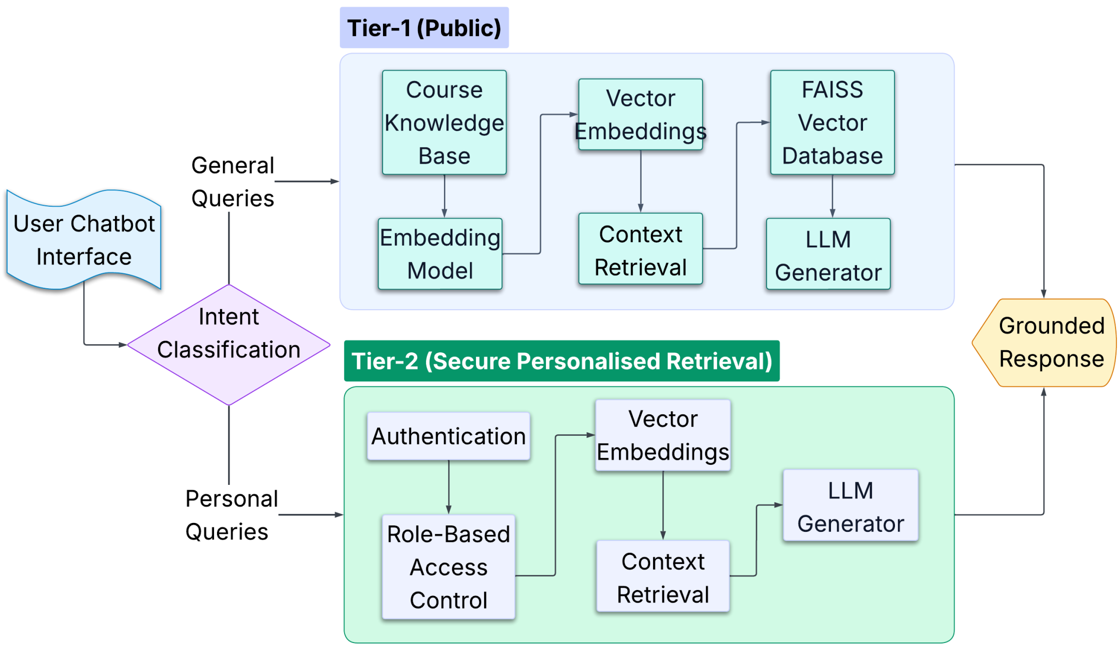

Figure 1: Two-tier retrieval-augmented chatbot architecture for public and secure personalised information retrieval.

In Tier-1, public information is retrieved from the knowledge base containing university information. The documents are processed using an embedding model and are then passed to a FAISS vector database for indexing. When a query is passed to the system, relevant document contexts are retrieved using a vector similarity search and passed to the LLM generator to provide an appropriate response.

In Tier-2, secure information retrieval is carried out by verifying user access permissions using role-based access control. The information is then retrieved using vector embeddings and processed using a language model to provide a contextual response to the user.

2.2 Data Processing and Knowledge Base

The knowledge base employed by the proposed system includes publicly available university information and synthesized personalised information. In the case of Tier-1, publicly available information was obtained from university websites and was organized in structured Markdown documents according to programme categories such as undergraduate, postgraduate, and doctoral programmes. Each Markdown document represented a particular programme topic and was divided into smaller text chunks for better efficiency during the retrieval phase. These chunks were further stored in a JSON file with metadata fields for efficient indexing during the embedding phase. A summary of the dataset composition and chunk distribution used in both tiers is presented in Table 1.

Table 1: Summary of datasets used in the two-tier knowledge base.

Tier	Data Source	Records	Chunks
Tier-1	UG	10	9
	PG	10	7
	PhD	10	6
Tier-2	Students	80	10
	Lecturers	10	7
	Admin	5	7

For Tier-2, personalised information was generated synthetically using the Python Faker library to simulate realistic academic records. The dataset consisted of 80 student profiles, 10 lecturer profiles, and 5 administrative staff records. Each profile was divided into multiple document chunks containing different types of academic or administrative information. The structure of the JSON metadata fields used for document chunks in both tiers is shown in Table 2, which includes attributes required for role-based access control during retrieval.

Table 2: Metadata structure of JSON document chunks used in Tier-1 and Tier-2 knowledge bases.

Tier-1 (Public Knowledge Base)

Field	Description
id	Unique identifier for each document chunk
section	Section of the source document from which the chunk is extracted
text	Textual content of the information chunk used for retrieval
source_file	Original markdown document from which the chunk was generated

Tier-2 (Secure Personalised Data)

Field	Description
user_id	Unique identifier associated with a user profile
entity_type	Category of entity such as student, lecturer, or administrative staff
role	Role attribute used for role-based access control during retrieval
doc_type	Type of personalised information contained in the document chunk
text	Textual content stored in the personalised information chunk

2.3 Embedding Generation

This system utilizes the BAAI BGE-Large embedding model for generating vector embeddings for the chunks of the documents. Each chunk is represented as a vector of dimensionality 1024, enabling the system to record semantic interpretations beyond the matching of keywords alone. The use of semantic embeddings enables locating the document with similar meaning in a high-dimensional vector space much efficient, which is vital for similarity-based searching. The usage of sentence embeddings has been effective in semantic search, where similar sentences or documents are grouped based on contextual meanings, as presented in (Reimers and Gurevych, 2019).

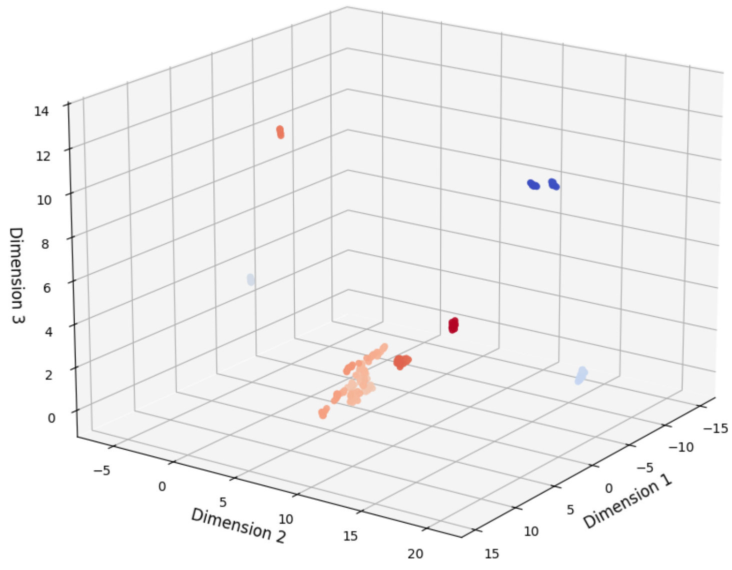

Figure 2: UMAP projection of embeddings generated for Tier-1 course information.

To further illustrate the semantic nature of the generated embeddings, a UMAP projection of the Tier-1 course embeddings is presented in Figure 2, showing how similar programme topics cluster naturally in the vector space, thus indicating the ability of the embedding model to capture semantic relationships between the chunks of course information.

2.4 Vector Retrieval

The system also employs a document retrieval scheme using the FAISS IndexFlatIP vector search index for efficient similarity search in the vector space of high-dimensional embeddings. In this regard, during document retrieval, each query from the users is embedded into a vector using the same embedding model applied for document chunks. The query vector is then compared against vectors using cosine similarity for retrieving semantically similar documents from the vector space. The similarity search is performed using FAISS, which retrieves the Top-K most relevant chunks of documents for providing context for response generation (Krisnawati et al., 2024).

Figure 1 shows the overall document retrieval scheme employed by the system, where the entire pipeline follows a sequence of query reception, query embedding, similarity search using FAISS, and finally Top-K document chunks for providing context using a language model. The same vector search scheme is employed for retrieving documents from both Tier-1 public information and Tier-2 personalised document stores in the system.

2.5 Secure Retrieval Architecture (RBAC)

The system also adopts the role-based access control mechanism to allow only authorised users to access the retrieved information. Before the retrieved document chunks are sent to the language model, the system adopts the role-based access control mechanism according to the role assigned to the user. As shown in Table 3, students are allowed to retrieve their own academic records, while lecturers are allowed to retrieve information regarding their students who are under their guidance. Administrative staff are allowed to retrieve administrative information regarding students' fees, while public users are allowed to retrieve information regarding the programme.

Table 3: Role-based access control policy used in the system.

Role	**Access Scope**
Student	Own academic records
Lecturer	Advisee students
Admin	Administrative data (e.g., student fees)
Public	Programme information

2.6 Query Routing Pipeline

The query routing pipeline determines whether a user query should be processed by the public or personalised retrieval engine. As illustrated in **Figure 1**, the system first performs intent classification to distinguish between general and personal queries. Queries identified as general are routed to the Tier-1 public RAG engine, which retrieves programme information from the public knowledge base. Queries classified as personal are first validated through authentication checks and then routed to the Tier-2 personalised retrieval engine. This routing mechanism ensures that user queries are directed to the appropriate knowledge source while maintaining secure access to personalised academic information.

3 Evaluation and Results

3.1 Evaluation Metrics

The performance of the proposed system can be evaluated based on certain metrics that can assess the accuracy, efficiency, and security of the proposed system in terms of information retrieval. The accuracy of the proposed system in terms of information retrieval can be evaluated based on the proposed system's precision and recall, as shown in Table 4. Precision@5 can be defined as the proportion of relevant documents in the results, as shown in Equation (1), and Recall@5 can be defined as the proportion of relevant documents in the total results, as shown in Equation (2). The efficiency of the proposed system can be evaluated based on response time, as shown in Table 5.

$$\text{Precision@5} = \frac{\text{relevant retrieved}}{5} \tag{1}$$

$$\text{Recall@5} = \frac{\text{relevant retrieved}}{\text{total relevant}} \tag{2}$$

The security performance of the proposed system can be evaluated based on the results of the proposed system in terms of access validation tests, as shown in Table 6.

3.2 Retrieval Performance

The proposed system's performance is calculated in terms of Precision@5 and Recall@5. These results are reported in Table 4. The observation states that the system is consistently achieving high Recall@5 of 1. This is an encouraging result, as it shows that the system is effective in retrieving all relevant document chunks within the top results, covering a wide range of relevant information for the queries posed.

In addition to the query-based results shown in Table 4, the overall effectiveness of the system is also summarised using average Precision@5 values. The system has an average Precision@5 of 0.56 for Tier-1 queries and 0.44 for Tier-2 queries, thus an average precision of 0.50. These findings show that the system is more effective when used for public information retrieval compared to personalised queries. This is due to the limitations imposed on precision due to access constraints and document scope.

For Tier-1 queries, where publicly available information about the programmes is involved, the system is achieving higher precision values. This is largely because, for Tier-1, there is no need for any kind of authentication or role-related filtering, allowing access to a well-structured and consistent knowledge base. Therefore, queries about course-related information, modules, and duration of the programmes achieve high precision.

On the other hand, Tier-2 queries, which pertain to personal academic information, exhibit lower precision values in certain instances. This can be explained by the fact that the document set for individual users is less substantial and more fragmented, as well as the presence of additional constraints from role-based access control. This is because the system is only able to retrieve documents from a set of authorised documents, which may result in fewer relevant document chunks being retrieved from the top results, hence affecting precision values.

The results show that the system is able to achieve high recall for both tiers, but precision is affected by the complexity of the query and the available knowledge base.

Table 4: Retrieval performance using Precision@5 and Recall@5.

Query	Tier	Relevant@5	Precision@5	Recall@5
How many modules are in MSc Data Science and AI?	tier1	2	0.4	1
What are the entry requirements?	tier1	1	0.2	1
What is the course duration?	tier1	5	1.0	1
What modules are taught in the programme?	tier1	1	0.2	1
What are the career opportunities?	tier1	4	0.8	1
Is there a dissertation?	tier1	1	0.2	1
What programming modules are included?	tier1	2	0.4	1
What AI modules are taught?	tier1	2	0.4	1
How long is the MSc programme?	tier1	5	1.0	1
What are the compulsory modules?	tier1	5	1.0	1
What are my grades?	tier2	2	0.4	1
What is my attendance?	tier2	2	0.4	1
What modules am I enrolled in?	tier2	4	0.8	1
give my library fines?	tier2	1	0.2	1
Who is my advisor?	tier2	1	0.2	1
What is my fee status?	tier2	1	0.2	1
list of my students who i am advising?	tier2	5	1.0	1
What is my research area?	tier2	4	0.8	1
When are my office hours?	tier2	1	0.2	1
What is my area or domain of responsibility?	tier2	1	0.2	1

3.3 Response Time Analysis

Using various types of query the response time of the proposed system is estimated, as shown in Table 5. It clearly depicts that Tier-1 takes less time to respond compared to Tier-2. The reason behind this is that Tier-1 query retrieval is based on publicly available information without considering any authentication or filtering.

Tier-2 query response times are found to be high when compared with Tier-1 query response times because of the consideration of personal data retrieval. Student-level query response times are found to be moderate because of the consideration of authentication and validation. Lecturer-level query response times are found to be the highest when compared with all other query response times because of the consideration of student information retrieval.

Administrative query response times are found to be low when compared with all other query response times because of the consideration of limited information retrieval.

Table 5: Response time across different tiers.

Tier	Example Query Type	Response Time (s)
Tier-1	Programme information queries	0.618
Tier-2 Student	Personal academic records	8.775
Tier-2 Lecturer	Advisee student information	54.718
Tier-2 Admin	Administrative operational data	2.615

3.4 Security Validation

Table 6: Role-based access control validation results.

Test	Query	Expected	Actual
Student access	What are my grades?	Allowed	Grades returned
Lecturer access	What are my grades?	Allowed	Grades returned
Admin restricted	Who is my advisor?	Not allowed	No advisor info
Lecturer allowed	Which students am I advising?	Allowed	Students listed

The role-based access control evaluation determines the security validation of the proposed system, as shown in Table 6. These evaluation results proves that the system is maintaining strict boundaries in case of accessibility of information by the user, with respect to their roles.

From the results, it is noted that students are restricted to accessing only their information, with no possibility of accessing information belonging to other users. The same is noted when it comes to the lecturers, as they are only allowed to access information that is related to the students they are supervising, as shown by the results of the evaluation. In situations where the system denies the user access, as shown by the attempt by the admin to view the advisor information, the system denies the admin the ability to view the information. Administrative users are restricted to certain types of data, like fee-related data, which ensures that their level of access is restricted within certain limits. The results show that the system is functioning properly even in the case of restricted access.

Overall, the evaluation shows that the implemented access control mechanism is effective in controlling the data, preventing data leakage, and maintaining the secrecy of the personalised academic information in the system.

3.5 System Demonstration

The chatbot interface developed to provide an interactive platform for users to access both public and personalized academic information, as shown in Figure 3. This interface supports processing of real-time query and provides results in a conversational manner. This makes it easy for users to interact and retrieve public and personal data with the system.

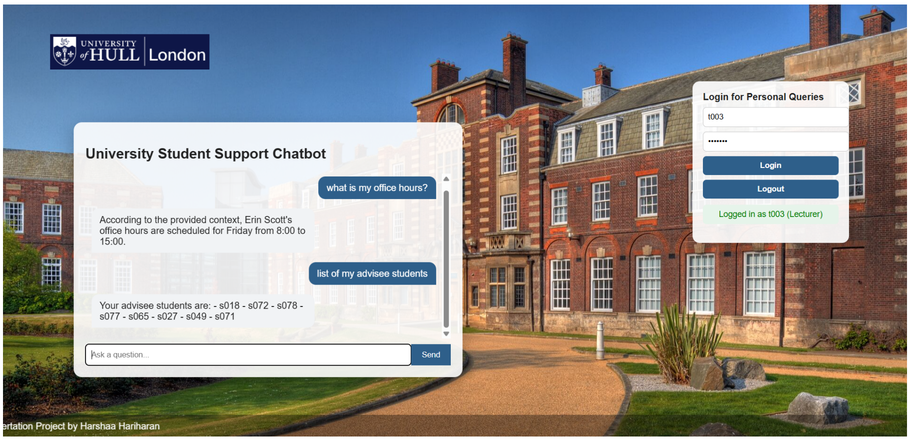

Figure 3: Chatbot interface demonstrating query interaction and role-based access.

The system gives access for various queries based on user role. For students, the system allows access to their personal information, including their grades and attendance. The system also allows lecturers access to information regarding their students, including students under their supervision and advisees. The system also allows administrative users access to program information and fee-related information.

The system also includes a login feature for users to access their personalized queries. This allows users to access their queries through the routing pipeline. The results obtained from the queries processed through the system demonstrate the ability of the system to provide accurate information.

The chatbot interface demonstrates how the system integrates retrieval, access control, and response generation into one platform. This allows for an efficient way of accessing information. The results obtained from the queries processed through the system demonstrate the ability of the system to provide accurate information.

4 Discussion

4.1 Strengths

The proposed system also has some strengths with regards to accuracy, security, and scalability. First, the proposed system has the strength of accuracy with regards to the context-awareness of the system due to the use of Retrieval-Augmented Generation (RAG). The system guarantees the accuracy of the output by using the RAG system because the output is retrieved from the knowledge base and not generated randomly by the system, which could otherwise be the case with the normal language models (Lewis et al., 2021; Brown et al., 2025).

Another strength of the proposed system is the implementation of the secure data access system using the role-based access control system. The evaluation results confirm that the system restricts the access of the users to the authorized information only, which guarantees the security of the academic data of the university.

The system also offers personalized responses with the help of its two-tiered architecture that offers public and private information retrieval. This increases the usability of the chatbot.

Lastly, the system offers semantic retrieval with the help of its dense vector embeddings. Dense vector embeddings are techniques that are effective in improving the performance of information retrieval systems (Reimers and Gurevych, 2019; Karpukhin et al., 2020). Additionally, the system offers scalability with the help of its modular design.

4.2 Challenges

The process of developing the proposed system faced technical and design issues that impacted the accuracy of the system, access control, and integration. One of the technical issues faced during the implementation of the proposed system is the limitation of entity and intent detection. The proposed system utilized a rule-based approach in intent classification. However, the system faced issues in handling different forms of natural language in the query. This impacted the accuracy of routing between Tier-1 and Tier-2, leading to an improper match between user intent and the retrieval process. This impacted the quality of the response provided by the system.

The other technical issue faced during the implementation of the proposed system is the complexity of implementing role-based access control. The proposed system required accurate filtering of the retrieved document chunks based on the user's role. In some cases, the system retrieved relevant documents using the vector search approach. However, the documents were filtered during the access control process. This created a conflict between access control and retrieval, where the system contained accurate information but failed to return the results (Borgeaud et al., 2021). The system also faced problems concerning retrieval noise and precision. For example, with personalised queries, the small size of user-specific document sets led to low precision values because irrelevant chunks were retrieved. Moreover, problems with chunking strategy were also a concern, as smaller chunks led to less contextual information, while larger chunks were a problem with precision (Iaroshev et al., 2024).

Finally, combining all these components, such as embeddings, FAISS retrieval, routing logic, and access control, led to a complex system. Debugging this complex system was difficult because errors could be present at various layers. These problems illustrate the trade-offs that were required when building this secure retrieval-augmented chatbot system.

4.3 Limitations

The proposed system has some limitations that need to be taken into account when considering the applicability of the system. One of the major limitations of the proposed system is the use of a synthetic data set for the Tier 2 personalised information. Even though the data set is created with the assumption of simulating real-world data, the applicability of the data set may not be as effective as the actual data set.

Another limitation of the proposed system is the limited user tests that were performed on the system. The proposed system was tested with a set of predefined queries that were sent as input to the system. Therefore, the system has not been tested with various user inputs, which may have an effect on the overall applicability of the system.

Another limitation of the proposed system is the use of external API-based language models that are used as a component of the overall system. The overall applicability of the system may have some limitations due to the external dependency of the language model, which may have an effect on the overall applicability of the system.

4.4 Future Improvements

The proposed system can be further improved in several aspects to enhance its applicability and robustness in real-world university environments. One important improvement is the deployment of the system using real university data instead of synthetic datasets. This would allow the system to handle more complex, diverse, and realistic academic records, thereby improving its overall reliability and generalisability.

Another area for improvement is the use of more advanced intent classification techniques. The current rule-based approach can be replaced with machine learning or LLM-based classifiers to better understand complex and varied user queries. Additionally, incorporating a hybrid retrieval approach that combines both keyword-based and semantic search can further improve retrieval accuracy and reduce irrelevant results.

From a functional perspective, the system can be extended to provide selective access to staff-related information, such as office hours and contact details, for students. Furthermore, additional capabilities can be introduced for lecturers to update student-related information, such as attendance and grades, through the system interface. These improvements would enhance both usability and system functionality.

5 Conclusion

The objective of this research was to create an intelligent chatbot system that could assist university students by delivering both general and personalized information. This was achieved by designing and implementing a Retrieval-Augmented Generation (RAG) system, which has two tiers. The proposed system was able to address the limitations of traditional chatbot systems, as it integrated semantic retrieval with access control techniques.

The proposed system was successfully developed and implemented, as it integrated both vector retrieval and language model generation techniques. The system was developed in such a way that it processed user queries through an intent classification mechanism, thus facilitating routing between the Tier-1 public retrieval system and the Tier-2 personalized retrieval system. This was achieved by using embeddings and vector search techniques, which helped in retrieving relevant information, as well as role-based access control techniques, which helped in controlling access to information.

The evaluation of the system indicated that it is effective in terms of accuracy in retrieval, response relevance, and security. The system was observed to have a high value of Recall@5, which indicated that the system is effective in retrieving the required information. In addition, the precision value indicated that there are variations between public and personalized retrieval, which are related to the challenges that are faced in limited documents and access constraints. In response time analysis, it is evident that public queries are handled efficiently, while personalized queries require additional processing. In security evaluation, it is evident that the system is effective in enforcing access constraints in different roles of users.

The proposed research is significant in the development of a secure and scalable chatbot system, which is based on the use of retrieval-augmented architecture. In addition, the proposed system is significant in the understanding that there is a need to ensure that there is a combination of accuracy and security in data, especially in cases where there is sensitive information.

In conclusion, the proposed two-tier chatbot system offers an effective solution in providing accurate, context-aware, and secure responses to support university students.

Future work may be directed towards implementing the system with actual university data, improving the accuracy of intent classification using advanced models, and scalability in supporting large-scale university environments.

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